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CET 3126C

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10/11/25

1. Definition of Computer Architecture

A computer consists of hardware and software that are joined and work together. This is what computer architecture is. It is the organization of components that make up a computer and their operations that allow it to function. There are three sections that make up the architecture of a computer. The system design is all of the hardware parts of the computer. This includes the data processors, memory controllers, multiprocessors, direct memory access, and the graphics processing unit, or GPU.

The instruction set architecture, or the ISA, includes the function and capabilities of the central processing unit, or the CPU. This is the embedded programming language and it is the software that makes the computer run. The microarchitecture is the third section and it defines the data processing and storage element and how they should be implemented into the ISA. This is the hardware implementation of how an ISA is implemented in a particular processor.

2. History Milestones

- 1834: Charles Babbage created the analysis engine, a mechanical machine that could be programmed by the user. It could store 1000 numbers of 50 digits each.
- 1939: John Atanasoff invented the first electronic computer, leading to the beginning of the digital age.
- 1940: Start of the first generation of computers using vacuum tubes. They were large, power-intensive, slow, and relied on machine language
- 1943: Tommy Flowers created one of the first digital computers, the Mark I Colossus. It was the first electronic calculating device, and it was used by the British to read German encrypted communications during World War 2.
- 1951: The first general purpose computer was designed by Eckert and Mauchly. This computer was geared toward business use.
- 1956: End of the first generation of computers, and the start of the second generation using transistors. They were smaller, faster, and relied on assembly language.
- 1957: The first successful high-level programming language was created, Fortran, by IBM to solve engineering and science problems.
- 1963/1964: The end of the second generation of computers, and the start of the third generation of computers using integrated circuits. They were compact, and introduced

keyboards and monitors. On January 1st of 1964, IBM announced their first computer with integrated circuits. The first computer to cover a wide range of operations.

- 1970: The UNIX OS was introduced
- 1971: The end of the third generation and the start of the fourth generation using microprocessors. They were now personal computers, GUIs, and networking.
- 1975: The first microcomputer ALTAIR is introduced, running on an intel 8080 processor.
- 1981: IBM releases the IBM pc, the first computer for practical use
- 1989: Microsoft releases windows for IBM computers
- 1990: The world wide web was introduced, and the first website from Tim Berners-Lee was created.
- 1998: Google was founded
- Present to the Future: We are still in the fourth generation, and are moving towards the fifth generation involving AI and nanotechnology. Eventually, we can reach the sixth generation which involves quantum computing, like qubits and superposition.

3. Classes of Computing Systems

Desktops prioritize user-friendly interfaces and general-purpose computing. Its typical constraints is that it comes at a moderate cost, and lacks portability. It has a high interactivity, powerful graphics for design and gaming, and is versatile for a broad range of applications. They can be used for video editing, graphic design, music production, video games, web browsing, and running office applications. Servers are for delivering services and sharing resources across a network. Their constraints are that they require high reliability, long uptimes, and being able to have secure data handling, resulting in requiring redundant power supplies and robust hardware.

They have fast data processing and retrieval, and high availability to serve multiple users and applications. They can be used for data storage, web hosting, and cloud services. Embedded systems are specialized for a single, dedicated function within a larger device. They have strict limitations on processing power, memory, power consumption, but allow for real-time responsiveness and have a high reliability to operate without human intervention. They can be found in home appliances, automotive systems, and consumer electronics.

4. Classic Components of a Computer

The five classic components of a computer are the input, output, memory, datapath, and control. The input is the data that comes into the processor from external input devices. This includes keyboards, mice, disks, and networks. The output is the data leaving from the processor to external output devices. This includes displays, printers, disk, and networks. The memory holds instructions, and most of them for currently executing programs. The datapath is key in manipulating the data coming through the processor.

It contains the program counter, or the PC, which holds the address for fetching instructions. It has multiplexers to route data through the datapath, and processing elements that compute new data values from old data values like in the arithmetic logic unit, the ALU, in simple processors. Programmable registers hold data, and can only hold a limited amount of data. There are also special-purpose registers that hold data needed for processor operations. The control generates control signals which direct the operation of memory and the datapath. It tells the memory whether to send or receive data, tells the ALU which operation to perform, as well as routing data between different parts of the datapath.

5. Transistor Scaling & Moore's Law

Moore's Law is the observation that the number of transistors on a microchip doubles approximately every two years, leading to an exponential increase in computing power. So as time progresses, devices will get faster, more efficient, and be more capable of handling more complex tasks. Dennard scaling states that the power density remains constant as transistors shrink. They are correlated because Dennard scaling explains why Moore's Law is possible. Smaller transistors use less power and generate less heat, enabling higher performance without increased energy consumption.

6. The Power Wall

The power wall is the physical limitations in designing computer chips, which prevent the continued, exponential increase in clock frequency due to unsustainable power consumption and heat generation. This meant the end of just increasing clock speeds as the main method to improve a microprocessor's performance. It emerged when the rules of Dennard scaling broke down. Transistors had gotten so small that the voltage could not be lowered proportionally, resulting in leakage current of the transistors that were not actively switching. As a result, this led to an increase in power consumption. Furthermore, as the clock frequency increased, so did the dynamic power, and therefore this leakage current resulted in more power being used. Overall, this led to an increase in heat output, ultimately arriving at this power wall.

Because the power and heat output of single-core clock speeds rose, it became too difficult and expensive to have cooling systems to dissipate the heat. This meant the clock speeds largely remained in the low gigahertz range. For thermal design, new cooling solutions had to be created like heat pipes and liquid cooling systems in order to manage the heat output of CPUs and GPUs. This also resulted in dynamic thermal management, where the chip will monitor its own temperature, and adjust its own performance to ensure it operates safely within its thermal design power.

7. From Uniprocessors to Multiprocessors

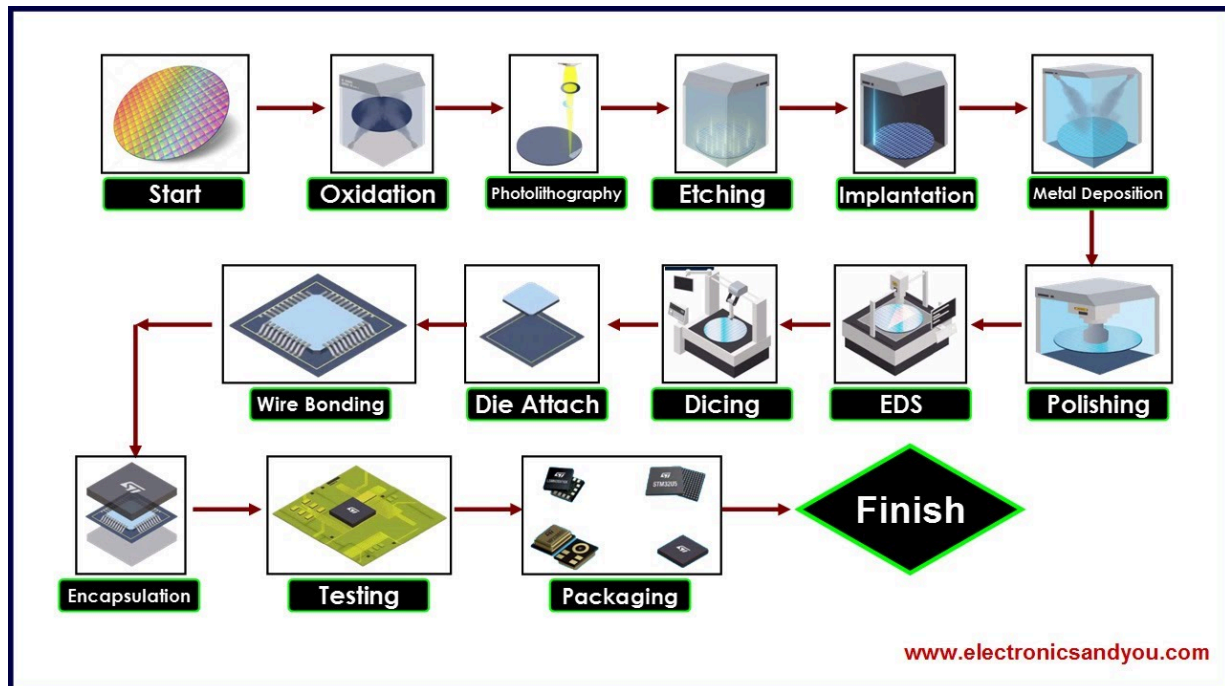
A multiprocessor, or a multicore processor, is a single integrated circuit that contains multiple core processing units, or cores. There was a shift from uniprocessors to multiprocessors for several reasons. Firstly, they are more energy efficient. They overcome the increased heat generation due to Moore's Law, and therefore decrease the need for cooling. This reduces power consumption, and increases battery life. Their performance is better because they can run multiple applications concurrently, and the decreased distance between cores on an IC enables shorter resource access latency and higher cache speeds. This increase of performance is dependent on the number of cores though.

For multiprocessors, true concurrency can be achieved. Aside from running multiple applications concurrently, specific applications can be allocated to different cores. Multicore processing actually increases the support for actual parallel processing within individual software applications. The way to increase performance was now to increase the number of cores, and not just simply the clock speed.

8. IC Manufacturing

There are several stages to IC manufacturing. The first process begins with a silicone wafer, where it is a bar or ingot of 99.99% pure silicon and is polished to extreme smoothness. Semiconducting materials are deposited on the wafer to enable the first layer to be printed on it. This process is known as deposition, and there have been advances in it, such as etch and lithography. Doping is the process of intentionally adding a small number of impurities to change its electrical conductivity. Lithography determines just how small the transistor on a chip can be. A lithography machine is used where the chip wafer is exposed to deep ultraviolet or extreme ultraviolet light. The light that is projected onto the wafer through the reticle, which holds the blueprint of the pattern to be printed.

To etch means to remove the degraded resist to reveal the intended pattern. During the etching process, the wafer is baked and developed. There are two types of etch, wet and dry. Dry etching uses gases to define the exposed pattern on the wafer, while wet etching uses chemical baths to wash the wafer. For the packaging, the function chip is enclosed in a protective package. There are several processes involved for the packaging, including die attachment, wire bonding, molding, and the final process of shipping.



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